A1.2 Nucleic acids (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 STATE the type of bond that links nucleotides in a strand, and the two carbon atoms it joins. [A1.2.11] [2]

2 NAME the two purine bases in DNA. [A1.2.12] [1]

3 STATE the number of histone proteins in the core of a nucleosome. [A1.2.13] [1]

4 STATE why radioactive sulfur labels protein but not DNA. [A1.2.14] [1]

5 STATE two of Chargaff's findings about the bases in DNA. [A1.2.15] [2]

6 DISTINGUISH between the template strand and the coding strand of a gene. [A1.2.11] [2]

Section B: Understanding

15 MARKS

7 EXPLAIN why the antiparallel arrangement of DNA strands affects how they are replicated. [A1.2.11] [3]

8 EXPLAIN why pairing a purine with a pyrimidine gives DNA a regular, stable structure. [A1.2.12] [3]

9 **EXPLAIN** how histone proteins hold on to DNA. [A1.2.13] [2]

10 **OUTLINE** two benefits of packing DNA into nucleosomes. [A1.2.13] [2]

11 **EXPLAIN** why many scientists in the 1940s expected protein to be the genetic material. [A1.2.14] [2]

12 **EXPLAIN** how Chargaff's data falsified the tetranucleotide hypothesis. [A1.2.15] [3]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

13 [A1.2.14]

Bacteria infected with labelled phages were shaken in a blender for different times, then centrifuged. The percentage of each label found in the liquid (outside the cells) was measured. The percentage of bacteria still alive was also recorded.

BLENDING TIME / MIN	0	1	2	4	8
^{35}S IN LIQUID / %	15	55	72	80	81
^{32}P IN LIQUID / %	10	18	20	21	21
BACTERIA ALIVE / %	100	100	99	98	97

a DESCRIBE the effect of blending time on the ^{35}S found in the liquid. [2]

b COMPARE the results for ^{35}S and ^{32}P after 8 minutes. [2]

c SUGGEST why about a fifth of the ^{32}P stayed in the liquid. [2]

d EXPLAIN why blending was needed, and why blending for longer than 4 minutes gave little benefit. [2]

e SUGGEST why the number of bacteria still alive was measured. [2]

Section D: Longer answer

6 MARKS

14 DISCUSS the evidence from Hershey and Chase, and from Chargaff, about the nature of the genetic material. [A1.2.14, A1.2.15] [6]